higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CONSTRUCTION PLANNING
NQF LEVEL 3

(12010253)

26 February 2019 (X-Paper)
09:00–13:00

REQUIREMENTS: A3 drawing sheet

Drawing instruments may be used.

This question paper consists of 5 pages and 1 diagram sheet.

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TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Answer QUESTION 2.2 and QUESTION 2.3 on the DRAWING SHEET.
5. Answer ALL other questions in the ANSWER BOOK.
6. Produce drawings, sketches and diagrams in pencil.
7. Sketches and/or diagrams must be neat, reasonably large, in proportion and fully labelled.
8. Rule off across the page on completion of each question.
9. Clearly cross out ALL work you do NOT want to be marked.
10. ALL abbreviations and symbols must comply with the latest National Building Regulations and ALL relevant SANS (SABS) codes.
11. Write neatly and legibly.

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QUESTION 1: COMMUNICATION IN THE CONSTRUCTION INDUSTRY

1.1 Differentiate between a *client* and an *architect* with regard to the importance of clear communication. (2 + 2) (4)

1.2 Give the advantages of each of the following examples of verbal communication:

1.2.1 Giving feedback

1.2.2 Monitoring

(2 × 3) (6)
[10]

QUESTION 2: TECHNICAL DRAWINGS

2.1 In the construction industry, drawings are an effective way of communicating ideas and visualising structures before it is built.

Explain each of the following forms of drawings:

2.1.1 Three-dimensional or 3D drawings

2.1.2 Two-dimensional or 2D drawings

(2 × 2) (4)

2.2 The DIAGRAM SHEET (attached) shows the front view of a round pipe.

Copy the given view and draw, to scale 1:1, the development of the round pipe next to the front view.

Use the following specifications:

- Diameter of pipe: 75
- Height of pipe: 100 mm
- Angle: 45° (12)

2.3 Draw, to an approximate scale of 1:10, a vertical section of a 270 mm cavity wall.

Show the following detail:

- Concrete strip foundation: 600 mm × 200 mm
- Concrete floor slab: 75 mm
- Screed: 20 mm
- Plaster: 15 mm
- Hard core
- Ground level
- Damp-proof course at floor-level height
- Wall tie
- Cavity filled with light concrete up to DPC level (13)

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2.4 Architectural drawings are part of technical drawings which are used to communicate construction information.

Briefly explain each of the following advantages of architectural drawings:

2.4.1 Shows clearly where a structure is situated

2.4.2 Used to estimate the cost of building a structure

(2 × 2) (4)

[33]

QUESTION 3: THE SOUTH AFRICAN CONSTRUCTION INDUSTRY

3.1 What is the difference between *trade organisations* and *regulatory bodies*?

(2 + 2) (4)

3.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (3.2.1–3.2.7) in the ANSWER BOOK. Correct the statement if it is FALSE.

3.2.1 The CIBD looks after the interests of homeowners and buyers.

3.2.2 The NHBRC controls the building industry by monitoring quality standards and providing technical support.

3.2.3 The ECSA is responsible for promoting the interest of the engineering profession.

3.2.4 The SACQSP ensures that high academic standards are achieved.

3.2.5 The MBA is a trade association that works closely with architects to ensure work is done within client budgets.

3.2.6 SAFCEC stands for the South African Federation of Engineering Construction.

3.2.7 The SACAP is a regulatory organisation set up in 2001 for South African architects.

(10)

3.3 Name TWO bodies or organisations that appear on both regulatory bodies and trade organisations. (2 × 2) (4)

3.4 Define the term *stakeholder*. (2)

3.5 Give FIVE examples of stakeholders. (5)

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3.6 Give the full name of each of the following trade organisations:

3.6.1 IOPSA

3.6.2 SAISC

3.6.3 ITC

3.6.4 CBA

(4 × 1) (4)
[29]

QUESTION 4: CONSTRUCTION LEGISLATION AND REGULATION

4.1 Explain the purpose of the National Building Regulation and Building Standards Act 103 of 1997. (3)

4.2 Name TWO ways to control the quality of a building with regard to acceptance criteria in the building industry. (2)

4.3 Briefly explain each of the following regulations in the construction industry:

4.3.1 Agreement certificate

4.3.2 Deemed-to-satisfy rules

4.3.3 Rational design

(3 × 2) (6)

4.4 The National Home Builders Registration Council is a section 21 company established in terms of the Housing Consumers Protection Measures Act of 1998.

Give FIVE functions of the NHBRC. (5 × 2) (10)

4.5 Differentiate between each of the following requirements in the construction industry:

4.5.1 Local bylaws

4.5.2 National Building Regulations

(2 × 2) (4)

4.6 Which tool, according to Building Regulation A19, must a competent person use to inspect a completed building before handing over the project? (1)

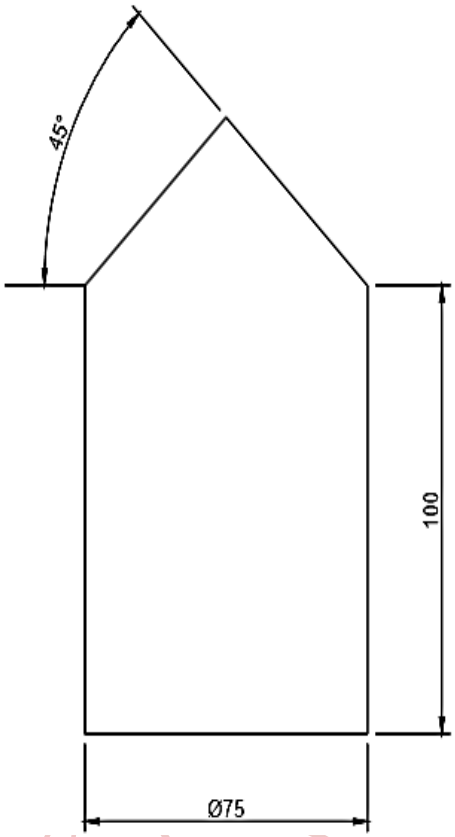
4.7 Define the tool mentioned in QUESTION 4.6. (2)
[28]

TOTAL: 100

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DIAGRAM SHEET

QUESTION 2.2



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